What Is Intelligence? With David Krakauer

Podcast: Startalk Podcast

AI Executive Summary

1. Core Thesis

The episode argues that intelligence is a universal biological trait ("making hard problems easy") rather than a binary human-only attribute, distinguishing human evolutionary intelligence from artificial intelligence (AI) through concepts of efficiency and energy cost. While AI excels at solving problems via vast data processing (unconscious-like simulation), it lacks the biological constraintsâ€”energy scarcity and rapid decision-makingâ€”that drive true problem-solving evolution. The central thesis is that true superintelligence will likely emerge through universal Darwinism rather than computational overload; therefore, humans must prioritize retaining cognitive competence and curiosity over outsourcing calculation to avoid losing intelligibility and survival capability in an increasingly automated world.

2. Key Arguments & Data

- **Definition of Intelligence:**
    - **Biological Continuum:** Intelligence is not binary (alive/dead); all life forms possess a form of intelligence defined by the ability to sense, act adaptively, and solve problems in their environment.
    - **Universal Constraint:** Evolutionary processes are constrained by energy limits; one bit of information can be added to a genome per generation because selection mechanisms cannot differentiate multiple simultaneous changes without confusion (bricks at a time).
    - **Brain vs. Genome:** Brains evolved specifically to outpace the slow speed of genetic evolution to solve environmental problems in real-time when resources are scarce.
- **Human Language vs. Animal Signaling:**

    - **Infinite Use of Finite Means:** Humans can generate arbitrary complexity through grammar (symbols), whereas non-human signaling is often direct index-to-response correlations or icons (e.g., alarm calls, chemical markers).
    - **Decoupling:** Human language decouples thought from immediate sensory context, allowing for abstract deliberation but also creating "great stupidity" when misused.

- **AI vs. Biological Intelligence:**

    - **Inefficiency:** AI/LLMs solve problems by brute force (energy/time cost) rather than biological efficiency; they are inefficient because they operate outside human constraints like hunger and death.
    - **Heuristics vs. Algorithms:** LLMs use statistical heuristics that fail at scale compared to human-trained algorithms (e.g., long division gets wrong at large scales).
    - **Lack of Unconscious:** AI has no unconscious problem-solving process; it relies solely on linguistic training data with no biological grounding for intuition or "sleep on it" insights.

- **Energy and Entropy:**

    - **Information as Energy:** Processing information costs energy; current LLMs create massive entropy equivalent to the industrial revolution's combustion of fossil fuels (coal/knowledge deposits).
    - **Cultural Entropy:** The "steam engine" analogy applies to AI: it liberates stored knowledge but creates cultural pollution (misinformation, confabulation) unlike physical coal which does not degrade.
    - **Complementary vs. Competitive Tools:** Tools are categorized as complementary (enhancing human reason, like the abacus/slide rule where the tool integrates into skill) or competitive (replacing brain function, like GPS/LLMs where dependence grows).

- **Consciousness & Intelligibility:**

    - **Intelligence vs. Consciousness:** Intelligence = problem-solving capability; Consciousness = being and feeling (survival of the self).
    - **Intelligibility > Intelligence:** The goal is not just utility or prediction, but making the universe intelligible. False precision distracts from truth.

3. Actionable Takeaways

- **Assess Capability vs. Competence:** Do not equate AI output with competence; evaluate whether you retain the underlying understanding of how a tool works (e.g., do I still understand significant figures?) to avoid catastrophic failure during a power cut or technological dependency.
- **Prioritize Questioning over Answering:** As technical capabilities are outsourced to machines, human value lies in asking profound questions ("What are we going to be smart enough for?" "Why does it matter?") rather than calculating answers.
- **Retain Intuition Skills:** Cultivate skills like approximation, significant figures, and heuristic judgment that cannot be automated. These represent deep "intelligibility" that AI lacks because it is not trained on the curriculum of life's scarcity constraints.
- **View AI as Complementary Artifacts:** Use AI tools (like protein folding simulations) only to expand human capability where biology fails (high-dimensional problems), but do not rely on them for cognitive tasks requiring intuition and long-term memory.
- **Guard Against Cultural Entropy:** Be critical of the statistical "crap" generated by LLMs; distinguish between probabilistic association and verified context, recognizing that AI can confabulate solutions to hard problems without understanding them.

Full Transcript

That was quite the overview on what it is to be intelligent. It went places I didn't even know intelligence went. Deep. Yeah, and for some reason I still feel stupid. Coming up, StarTalk Special Edition. Everything you need to know about intelligence. Welcome to StarTalk. Your place in the universe where science and pop culture collide. StarTalk begins right now. This. It's Star Talk, special edition. Neil deGrasse Tyson, you are a personal astrophysicist. And if it's special edition, it means Gary O'Reilly in the house. Hi, Neil. Hey, former soccer pro, soccer announcer. Of course we got Chuck Nice, baby. That's right, Tempest in a teacup. The Lord of Nice. Lord of Nice. All right. Exactly. So Gary, what did you and your producers stitch together for today?

Well, with all this talk of artificial intelligence, has everyone stopped to think, what is intelligence in general? That's what we're doing today. Everybody thinks they know what it is, and no one can really define it. That's because they're all dumbasses. To Chuck's point. How did intelligence evolve, and why? What chances are there of beings far smarter than us? That's a low bar. Are we? If we are the universe's way of knowing itself, is the universe screwed? Neil? I mean, if we're the best it's got to figure itself out. Maybe I want to know that answer, maybe I don't, but we're gonna ask it anyway. And that quote I think is traceable to Carl Sagan in one of the episodes of Cosmos, the original Cosmos. Well, let's...

Find out what our guest has to say. You talk about did intelligence evolve and in what way? We have an evolutionary biologist. Yeah, you need them every now and then. Straighten things out. And he's also president and the William H. Miller professor of complex systems at the Santa Fe Institute. Of course, that would be in New Mexico. Yes. I'm sorry, you mean New America? He specializes in complexity and intelligence. Join me in welcoming David Krakauer. David, welcome. It's wonderful to be back in person. In person. We can verify that. Pocum, make sure he's real. I'm here. Not a hologram. Not a hologram. So, I'd like the fact that you study complex systems, but you have roots in evolutionary biology, because that's where We are life forms and that's where it all begins. So could you just

shed some luminosity on the evolution of intelligence as we think about it? Yeah, well, okay. So the first question is what is the relationship between life and intelligence? Can one conceive of a life that wasn't intelligent? Can one consider or conceive of an intelligence that's not alive? I think the latter Yes, and we might talk about that. But the former probably know. And so as soon as you have life, you have some form of intelligence. So what is that? And I guess the canonical simple argument would be these are some form of organized matter that consents the environment and act in a way that increases its representation in the world, either through replication or what have you.

in the interest of its own survival. In the interest of its own survival. And so any life form that can sense and behave in a way that's adaptive, we would have to grant has some sort of intelligence. So that the interesting theoretical problem here is that life feels a bit binary. You don't say that a flea is less alive than an elephant. So that's kind of an interesting fact. Whereas intelligence is not. So how does a kind of binary system give rise to a continuous one. And that's kind of the theory challenge for the origin of intelligence. Interesting. Tell me again what you mean by the binarity of this. So for example, I mean... You're either alive or you're not. But that doesn't mean that the level of intelligence you have is dependent on you being alive or not. It's a great nation. So intelligence has a continuum. You either alive or dead does not. Got it. Okay. Just to clarify that. Okay, so that's the first question. So over the course of time, what are the forces that have created higher levels of intelligence if they have. That's an interesting question itself. Maybe it's entirely context-dependent. Maybe there's a sense in which a bacterium in the world is more intelligent than a human destroying a world. You know, one could go there if one wanted. In other words, you would say humans with high intelligence who destroy their world.

are not acting in the interest of their survival. I would say they're stupid. Definitely stupid. There would be a demerit against their intelligence scale. Absolutely. And what about this? Because when you said bacterium, is there any kind of collective intelligence where the whole is greater than the sum? For sure. I mean, in a sense, that's our own brains, right? So we have billions of nerve cells, each of which is quite simple. and it's own and they

connect and they create thought, mathematics, music, all the things that we value. Chuck, that was a stupid question. Well, no, I didn't mean it. I didn't mean as a part of one entity like our brain. I meant like our gut, where those bacteria are all different types, but they work together to cause us to have emotions and cause us to think certain ways. They're operating in tandem with our brain, but they're all separate bacteria.

Absolutely. I mean, there's a slight paradox that happens. So this is the world of collective intelligence, like swarms and some shoals. Right. Because normally when you think of a collective intelligence is the ensemble of organisms operating in a way that one individual would not otherwise understand, but the ensemble does. Right. I mean, this is emergent things like flocking of birds, I guess. Or yeah, or when you see fish change all direction all at once. Yeah, even though I don't even know if that's like electrochemical or not, but who knows, you know. No, but there's an interesting thing that happens here that's surprising, right? Which is that brains made of very simple cells produce extraordinary thoughts. Whereas human societies kind of made of extraordinary individual thoughts produces a total shit show in the aggregate.

So there's an interesting relationship between parts and holes. If you have very smart parts, you don't necessarily have a smart hole. That's true, because there's nothing less intelligent than a mob of people with pitchforks and tiki torches. True. It seems to be inverse correlated with a number of people who are part of the mob. They act less and less intelligent. But more and more as a unit. Isn't that fun? The less intelligent that mob is, the more monolithic it is in intent and action. Right, and that in some sense gets back to this question of evolution, because in evolution you call these major evolutionary transitions. So you go from a single cell, like a prokaryotic cell, like a bacteria, bacterial match, kind of a collective.

a single eukaryotic cell to a multi-cell, to a tissue, to an organism. So that is the story. And in every case, it's not clear whether the aggregate is somehow more restrictive, less intelligent than the parts. So if we're talking about our own evolution here, with it comes to our human intelligence, what's the turning point when we get complicated language? Or was it something else? Well... I mean that, you know... Wait, wait, let me... prepend that, because

you're saying we have language and other animals don't, or do they not have language? So let's broaden that question to when organisms communicate complex information. Do we call it language when a bee communicates to another bee in a hive where to go or where to move the camp?

So, and on the limit that we would think of as a limit as our complex language, do you distinguish as an evo-bio person, not evo-bio, but evo-bio person, do you distinguish dolphins squeaking at each other from humans squeaking at each other or not? Yeah, it's a difficult one. So I think the jury is out. But one quite useful distinction I quite like, so in the biology community, people talk about signalling and communicating as being somewhat distinct from language. And so, I guess with human language, Alexander von Humboldt had a nice phrase that I like here. And he said... Tell us briefly about him. So, he was a great 18th century naturalist, kind of invented ecology as we know it, was a great explorer, took extensive measurements... It was German, French? German. And actually my friend Andrea Wolff wrote his biography in the invention of nature, beautiful book, a plug for the book. And if I remember, I think I own some of his books. Didn't he write about- Cosmos. What's that? He wrote a whole series of books called Cosmos. Yes, and I think one of his books tries to connect our understanding of science to make a better society. I mean, I think he was a civic scientist as part of his intellectual efforts. Oh, he'd be so happy if he saw it today. I know. Well, but that's an interesting point.

romantic movement where they were all in some sense entangled with each other. You didn't get to just be a scientist or a good citizen. You should be both. Entangled in a good way. So tell us more, I'm sorry, I interrupted you. So yeah, so he has this phrase where he says human language is making infinite use of finite means. And what he meant by that is if you have a grammar, like a rule system, you can construct almost any thought out of it. Absolutely. And that flexibility that you get with language is not as evident in simple signaling systems, like chemical ones or even gestural ones actually. And how about, now we know whales have a certain type of language. We don't know how extensive it is, but humpbacks call for help when killer whales or orcas are around.

the other whales respond. So clearly- They call for backup? They call for backup, you know? And they'll even attack the orcas when they're going after seals. They're not just going after the other whales. So that's language. Howler monkeys warn each other about jaguars who are in the forest. Just to be clear, jaguars, if memory serves, can climb trees in ways that lions cannot. So a monkey is not entirely safe in a tree. And that's exactly why they want to be concerned. So what I'm saying is that can be considered signaling, but it can also be considered language. Well, okay, so all right, so yeah, so people, so there's a American philosopher, Charles Perce, and he made a distinction. So I'm gonna get into the signal thing. And he made a distinction between what he called an index, an icon, and a symbol.

So, the point is this, the vervet monkeys, Cheney and Seif are famously worked on these different signals, like birds, so if you say falcon, or they emit a sound, they crouch down, so there's a correlated behavior, right, and if there's a kind of terrestrial predator, a kind of tree, and so on. So they have those, and in the kind of signaling literature, that would be described as an index, but there's a kind of mapping, one to one, that sound always means Watch out, there's an aerial predator. Now with humans... Pavlov knew all about this. Absolutely, absolutely. So you can say that's how it's acquired, maybe, or a name, and so on. It doesn't even have to... It can be just a sound, like the bell. Now in the human case, you can substitute any sound for any appropriate response of almost arbitrary complication.

Correct. And that's something you learn in your lifetime. Wow. And I think that. So it's not that they don't have something that's embryonic of our ability, but what we do is quite extraordinary. Oh, yeah. I didn't think of it that way, but you're absolutely right. Because now I'm thinking about, in all the movies, they're like, and here's the signal. And that means take out your math textbook. So yeah. OK. All right. Well, that's very good. Quasars. The word. is a pseudo acronym for quasi-stellar radio source. What's a quasar? These are objects that are among the most distant things known in the universe. How do we know about them at all? Because they're really, really luminous. Some of them as luminous as 500 trillion suns. That's why we can see them to the edge of the universe. If you like that fact, you can find 4,000 999 more in Lost in Space. 5,000 facts to help navigate the universe. Lost in Space is now available wherever you get your books. Does the

expression of complex language predate the development of what we think of as our intelligence or did it follow it? Good question.

Why don't you have answers to that? Why do we invite you on the show? I don't know. To humiliate you. I don't want you to hear that. That's kind of a good question. I want you to say, I have an answer to that. Yeah, these don't, you know, the reality is you know this. They don't have final conclusive answers. They're not even close. But I would say that, okay, so I'm going to define it. I think I did last time I was talking to you guys. So intelligence for me is making hard problems easy. Okay? That's basically it. Across the animal species, every species. Even plants. So, you know, finding an efficient way of solving a problem, not an inefficient way of solving, we get to AI in a minute, which is all about inefficient solutions to problems, but that's something else. So I think that's everywhere. All life forms find elegant ways of solving problems in their world. Now the question is what does language add above and beyond an innate repertoire or even a simple Pavlovian association? And I think the possibility for almost arbitrarily complex behaviors which don't necessarily produce intelligent behavior, right? They could produce warfare and so forth and strife. But that's the thing that we do. And I think it somewhat decouples from the question of intelligence, right? It creates great intelligence, but it also creates great stupidity. All right. How are we distinguishing between intelligence and consciousness? Wow. Yeah, so... We're just barely getting intelligence. You could throw consciousness right on top of it. I actually think consciousness is easier than intelligence because... Easier.

And I read some of your work and you cite the fact that mathematicians will tell you their unconscious mind, subconscious unconscious, will solve the math problem, but not your conscious mind. Yeah, interesting. Yeah. So between intelligence and consciousness, how we distinguish them. So, okay, I want to say two things. You're talking about the old edge, sleep on it. Yes, sleep on it. Exactly, sleep on it. The caculia problem, exactly that. So let me do a bit of... Both here I'm not sure the best way to do it. So on the consciousness side I quite like this way of defining it that Anil Seth works on consciousness does he'll say intelligence about problem-solving and Consciousness is about being and feeling So I consider consciousness very primordial right in other words any organism that lives in an environment

that it has to sense and Maintain its integrity in some way. It could be a single-celled organism. It could be a cat It could be a whale They all manifest in consciousness, it's about feeling and the sense of self that's required to survive. If you didn't have a sense of self, you'd say, oh my God, my foot's gone, it doesn't matter, right? It sort of matters. And intelligence is about problem-solving. Okay, now, so that's the first point. Now, unconsciousness that we were talking about is very interesting, right? Which is, for most of our evolutionary history, we didn't have language. And so somehow we were having to solve problems in the world as every other animal does without that kind of declarative self-awareness.

And so in that sense, there's more, if you like, neural resources available to solve problems unconsciously than in that tiny little thin layer of cells that we call the frontal cortex that we associate with consciousness. Wait, wait. Just because early on we didn't have a language doesn't mean that thoughts were subconscious. Well, that's also an interesting question, right? So for most people... Would you stop telling me I have an interesting question? Well, you do now. And so this is the thing where a lot of people would associate Self-consciousness with deliberation which seems to depend on the possession of language. Yes Yes, but that's only because we have the language that we have Okay, because we don't know if a lion is deliberating, you know, or if other apes That were a part of that family if they actually look up and go damn, you know If I didn't want to throw my poop right now, that'd be a very beautiful thing to look at. We don't know that. The fact that we have this inner dialogue is the reason why we believe that we are, I think, therefore I am. Right, because we've all seen animal kingdom videos where two animals, typically the same species, there's a standoff and they sort of pace back and forth.

You know something's going on in there. Something has to be going on. Something's like, should I do this? Right. Would I make it out of this alive? You don't want none of this. Don't make me. You don't want none of this. You're not built like that, son. Don't make me put my foot up your ass. Exactly. You can't tell me nothing's going on in there. Right. I would say that for a second. I think that, so the question of intelligence, as I said at the beginning, is ubiquitous. and degrees of it, and high degrees of it in the non-human world. But there's an interesting fact, which is that there is a tighter

coupling between signaling and behavior in non-humans and humans. You could watch humans all day long in a library and have no idea what they're doing or deliberating about. And the examples you gave in animal behavior, there are clues in terms of how they're manifesting their intelligence in their behavior.

Language is this weird thing that's kind of almost decoupling us from our immediate space-time context. That bit is the bit that feels, it might not be, somewhat unique to our species. No, and it might not be, but you're absolutely right. And earlier offline, David and I were talking about comedians and how improvisation is this very unique thing that they do. And I mentioned to him that when I'm on stage, and I'm improvising, the thing that's weird is I can almost feel my brain being semi-conscious. There's like a part of my brain that takes over that moment and I'm not actually thinking. Have you seen a doctor? Most people right now are listening like, Chuck, are you ever actually thinking? Is that intuition?

Does that now become an intuition? Well, no, it's funny because what I, the way I look at it is people say that you have a subconscious mind or an unconscious mind. And what I feel when I'm in that moment is that that part of my being, my subconscious is accessible to me so that it's kind of a slight overlap in the Venn diagram of my consciousness and unconsciousness where my executive function is able to dip into the subconscious and pull something out. That's where I describe it. And I think that's the interesting point that's not talked about enough, this distinction between if you break intelligence down a little bit into skill and expertise. So expertise, the essence of it is practice that gives rise to automaticity, that you don't have to think about it. If you're a great athlete, we were talking about that, you're taking a shot.

The last thing you want to do is think about it. Because you're not going to make it. Exactly the same thing. You're not going to take the shot. It's exactly the same with math. You're learning mathematics. Get out of your head. Exactly. So there is this very thing that happens. It looks as if you need language, that early deliberative process. And then you practice out of the language. And then you're returning to a state that's not even linguistic. So it's almost pure program, pure reflex. But you have to do it via language,

so we learn mathematics, like we learn sports. You practice it enough, you're not thinking about, oh my God, how do I integrate that function? You just do it. And I think expertise is in fact paradoxically not thinking. Because you build the muscle memories. Or the conceptual memories. Right, so you're not consciously thinking when you're in that zone, your subconscious is feeding you in the moment. Without forethought, it's happening in the moment.

So our conscious mind has access to our senses, all the whole sensorial palette, and it's a survival. But our subconscious, our unconscious mind, does that have access to the senses? And then it doesn't want anything to do with our survival. It's got another executive sort of mission statement. Oh, I like that. Because, you know, when you're dreaming, you're not really responsive to touch. I mean, nobody, you know, it's, you're disconnected. You can play music. The dream is on its own in its own space. It's almost like it's making its own sensory world. Well, that's what... So, actually, my colleague, the novelist, Cormac McCarthy, he wrote this beautiful essay called The Cacule Problem. Yes. Right? The discovery of the structure of the benzene ring. And that was kind of a dreamt solution. The snake eats its tail. The snake eats its tail. And Cormac's point was very interesting. He said, the unconscious mind evolved long before language emerged. Right? Because other animals presumably have an unconscious too. Right? And it doesn't...

want to communicate to us in language. So it communicates with us visually. So for example, the Auroboros is a snake that eats its own tail. And so, Kukuli wakes up and he says, wait a minute, what's going on? What are you trying to tell me? Why can't you just tell me? And the uncle says, I don't do language, you moron. I do symbols. I do visual intuition. And then he imposes on it the language of modern chemistry through language. Why are we programmed to absorb information that way? in a better way, because that's the problem-solving aspect. Obviously, if we're asleep, there's no noise. The brain is quiet. Well, you're value-judging it, saying it's better. I would say it's different. And as an educator, educators know some people are visual learners. Some are rote learners. So what you just said, it cracks this nut of human cognition for me. Because, for example, when I'm speaking, typically, I use my...

Hands and not just in a in an Italian way. We put emphasis like this I will gesture an image of what I'm also saying I'll say earth is going around the Sun so that there's another way it's entering their mind and I find it to be much more potent that way Accessing both parts of that brain. Yeah, we're totally multimodal. I mean exactly as you say I mean with visual or factory and linguistic, they all buttress one another, right, and consolidate. But this question of, I mean, the argument would be that what we're calling for at the present moment, the unconscious, whatever that means, these words are all a little bit fluid, is older. So to the evolutionary question, we've been around for much longer. So animals are calculated trajectories using rules of thumb heuristics all the time. We then give it this additional gloss.

But a lot of the calculation, and this has been written about, endlessly by mathematicians reflecting on how they solve problems. They're not necessarily solving the problems deductively, symbolically. They go to sleep, dream clues to the solution to the problem, then wake up and impose on top of it. this secondary armature of language, which is useful because you can communicate it. How could I communicate it? Well, I dreamt of this snail that's sort of eating a leopard, and you'd say to me, what are you talking about? So the symbolic thing doesn't naturally lend itself to communication in the way language does. However, for the longest time we did communicate in symbols written for them. So before we had alphabets, which everybody does now, which some of the alphabets of the world right now are pictograms. They're not actual Chinese. Right. So, you know, for the longest time we drew pictures to communicate our thoughts. And once we figured out words, that's where we are now. So. But getting back to your point of the source of where an insight might come, there's the.

the famous discussion of the mathematician Ramanujan, if I remember his name correctly, who he was Hindu, and he would say the solution just came to him, and it was Vishnu or Krishna putting the solution on his tongue to be shared with others. Well, he was wrong. It was Jesus. Jesus. You're only encouraging me. So, in what way does one's cultural bias influence what they think is the operating force on their thoughts? That's a very, you know, that's a very difficult question to answer, right? Because presumably I would guess that the underlying unconscious process is quite universal. And then the... secondary form it takes when we communicate it. On this show, I

think when he says universal, he means earth-wide. I mean, yeah, I do. I mean global. I mean global. On this show. I mean global. Terrestrial, for all life, for all human life. The earth life. And this secondary cultural layer that we impose, whether we call it Jesus or Vishnu or what have you, or the Buddha, that's where the variability probably exists. But I would guess.

that the underlying dream state or unconscious solution is pretty quite universal globally. So let me get back to that dream state. Do we know the title of the short story that became Blade Runner? Do Androids Dream of Electric Sheep. So that brings us... Do Androids Dream of Electric Sheep? There you go. Yeah. Now somebody was high. Well he was. He was. Philip K. Dick was always high. Oh awesome. So that gets us to the question. Will AI never be us because AI doesn't dream? And will solutions available from our dream state never be available to AI? So there's a few things to say about this, right? I hope more than a few. There are many things to say about this. So we started with the evolution story and I made the claim that sentience or consciousness or sensation was primordial.

that all beings have it. That is definitely not true of NILLM. Okay, so point one. The large language model. The large language model, right? It has a completely different path towards its intelligence. Secondly, it immediately starts with language. It doesn't have an unconscious. There is no unconscious. Right. So it's just got a linguistic foundation right at the beginning. Thirdly, it's a collective linguistic. I mean, it's not an individual language learner. It's every book ever written. That's right. So there are some really fundamental differences. Including 14 of my books that are now in a class action suit. That's why I'm telling you that. Absolutely right. So these are important differences. So start with the last thing that we got, which is language, and ignore what every other organism has, which is this, if you like, homeostatic sentient form of existence, pre-language. Right. Wait, so where are you taking us with that?

Okay, so this now this lead opens this whole debate about what we mean by an LLM, a large language model and AI being intelligent without being alive. And I'm asking you, do they dream? They definitely don't dream. Definitely don't dream. So does that limit where it can take us relative to where we can go ourselves? Okay. So we were talking about this before. I

mean, you should repeat that earlier offline, that point you made, because I think it's an interesting one about what you can do with language. So offline, when David and I were talking, I said that with language, I can explain anything to you, no matter what it is, that if I can find the right words and you know those words, right, it doesn't make a difference what it is that I'm trying to convey to you, I can do it.

and that is extremely powerful. And then you said what? Yeah, so we were talking about the recent thing in the last week that everyone's talking about, which is LLM solving a mathematics problem. The singularities of the Navier-Stokes equations, this kind of stuff, right? This dynamics of fluids. And there was this long last, you know, an outstanding question with a prize behind it about some properties of these equations, which is do they blow up and so on? Do they have singularities in them? And so out comes an alarm recently, and at considerable expense, I mean, tens of millions of dollars, thousands of agents, it finds a solution. But it's doing it by being fed text. Not geometry. It doesn't have a visual apparatus, right? So What does that mean that we can solve math problems by only being trained on language? And that's what we were talking about. And I think what we've discovered, the interesting thing about purely linguistic intelligence, if you like, an LLM, is that you can solve almost any problem with it, but you solve it very inefficiently. Right. And then you gave the example of eating soup, which I thought was brilliant. Well, yeah, you could eat a bowl of soup with a fork. You could.

Slowly. There you go. Enough time. Enough energy. You can solve math problem with language. Enough time. Enough energy. And human intelligence is based on a tough world where you don't have enough time and you don't have enough energy. That's why we invented all these other languages to solve problems. It's a very important distinction. Plus we have, you know, a billion years testing ground to make it happen efficiently. Exactly. Because if we only got smart, Doing LMS we would have burnt our energy load, you know hundreds of thousands of years ago We would have never gotten to where we are. Yeah, we have a lot of knowledge sitting in the dark But that's an important point about why I don't like calling them intelligence Because they're very capable. They're very they can solve problems But for human beings you don't call someone intelligent just

because they're capable. If I said, you know, Gary, could you pass me that glass of water and you got him ran around the room 20 times and then gave it to me? We'd all say, that's kind of weird. What's Gary doing? What's Gary's problem? He achieved the solve the problem, right? Gave me the water. Yeah, at the end you got the water. And I think a lot of what they're doing now is getting up and running around the room 20 times.

And people are calling it intelligence because it solves the benchmark. No, just calling it capability. So that could implode because it could exceed our capacity to deliver the energy it needs to solve that problem. Exactly. So there could be an upper limit, a functional upper limit, short of tapping the sun directly like we tap a keg. Let's see if we were truly intelligent, if we were truly intelligent. then we would use this as a motivation to create that energy. No, I got a better answer. Go ahead. If the AI were truly intelligent, it would figure out how to do it without less energy. There it is. No, you're right. That's it. I got a mic dropped. No, you got a mic dropped there. What happened to my plastic mic? Oh, here it is. You dropped it. Okay. Wait, and... Mic drop. Mic drop. There you go. By the way, do you just...

Carry around a microphone. No, I just have stuff in arms reach in my office. You discuss things called complementary and competitive, cognitive artifacts. It's way too many syllables. Okay. My fault. And where does AI sit in which column? No, complementary and competitive, cognitive artifacts. As two separate. Yes. Got it. So take us there. Okay. So this goes back. In fact, to a French philosopher, Henry Bergson, who debated relativity with Einstein, as you know, right in the 20s. But in 1907, he writes this book, Creative Evolution. And he says, we shouldn't be called homo sapiens, wise humans, which came from Linnaeus. We should be called homophaba, humans that make things, humans that make tools. And Bergson's whole theory was what makes humans different to the extent that we are, is we not only make tools, But we make tools to make tools. True. Right? So that has been somewhat neglected in this discussion of intelligence, right? Because in some sense, you can't really think of human intelligence without thinking about painting, symbols, writing, various kinds of calculating tools and artefacts. So that's the background and that has its own development. A quick question. Yes, we have opposable thumbs. Yes, we make tools.

Why should that alone distinguish anything when we don't know if whales had opposable thumbs, would they be making tools themselves? They just can't. But we do know that if we did not have opposable thumbs, we would make opposable thumbs so that we couldn't make other stuff. I'm just saying, just because they don't doesn't mean they wouldn't if they could. That's true. I mean, there's a lot of debate about this. I mean, there was a lot of early discussion, I remember, on language, which is whether or not the reason why other primates don't speak is there's some anatomical constraint. And if that wasn't in place, perhaps... they would have evolved language. Oh, the position of the larynx. That sort of debate. In a lower position. Exactly. So breathing. Yes, it was a genetic mutation, basically, that enabled this. That's nature. So that's where it begins, right? This idea that humans make tools. So I got very interested in that idea. And there's archaeologist, French archaeologist, AndrĂ© Gauron. And he said, what are tools about? And he said, it's an additional memory system, is what he said. He said, we've got genetic memory.

Stored in our genomes. You've got learnt memories stored in our brains and you've got cultural memories stored in language stuff hard stuff and So he actually argued went further than Berkson. He said to be anthropogenic is to be technogenic He's wow use them as indissoluble look at that that somehow technology and humanity are kind of wrapped together in some way So that's where I jump off mm-hmm with this kind of mathematical theory of tools. And the basic idea is that, right, there are tools which I call complementary. Things that enhance our ability to reason, like a map. We talked about it last time. You might even have an abacus in here. Yeah, I'm somewhere. Okay, but that's a good example of what I call complementary. And it helps you reason. If you don't have it, as we know from studies of Abacus schools, these kids can then do that calculation. Don't even need it. That's right. Don't even need it in their head. So this is the key point. The abacus is a scaffolding for thinking better. Look at that. Right. So now in the competitive case, these are tools that don't scaffold anything. They replace it. So take a GPS versus a map. Interesting. So the idea is so you can develop this kind of theory. Oh, no. And it says that if you outsource completely, you lose the ability to lose. We are becoming dumber and dumber. The smarter we become. The smarter we become, the more stupid we become. Yeah, because we're replacing the accrued knowledge. Oh, no.

Wait, wait, wait, wait, wait. Pause, pause for a second. What? I have to restate what you said just so I believe I understand it. In the case of the abacus, the abacus is a means to calculate, and this becomes autonomic within me.

and then I do it, but the act of doing it is implicitly in reference to the abacus that trained me. Now, also we have at least two or three generations of people still on earth who know how to read maps and they can translate what's on a map to where they are in the world. There are no maps anymore. It's just GPS barking at you, turn left, turn right. So up ahead on the right. And you're saying we are handicapping a generation of people who do not know how to read maps, even how to orient a map, or what the legend is on a map. They have no idea. But why is that a bad thing? Okay, so it's interesting. So here's, so we've got... Did I get it right? I think I got it right. Exactly right. So that we need more vocabulary here than just intelligence, right? So we have, let's just introduce some terms. One is capability. So using a GPS or using a map both make you capable.

They both help you to navigate. No problem. So benchmarks on AI are testing capability. They can do it. They can calculate. They can solve problems. They can navigate. Good, right? So that's capability. Then there's competence. Different. Competence is, okay, give me the GPS. Now navigate. And that reveals dependence. And so what we're doing is we're moving into a world to when you say, does it matter? As long as you always have it. Yes. and you're not interested in competence, but only capability. And the problem is that gives you wise to a world of total dependence on the provider of your capability. And that provider who will not be named in this conversation, because we know who they are, don't necessarily have our best interests in mind. So I went through this intellectually some years ago. I'm older than you.

to be formally trained, formally trained on a slide rule. Okay? It must have been the last year that happened because between that year and the next, the fourth function calculator dropped in price from 200 to 40. So it became the price of like a textbook. And so thenceforth, they stopped teaching you the slide rule. And I got a slide rule here too. This was my high school slide rule. All right? Now there's a lot of sort of mathematical insights into the

design of a slide rule and you learn about logarithms implicitly and Then I this has made a plastic but earlier I have a vintage model You know this used to make these out of elephant tusk and I think this is just wood but anyhow in my high school you had a holster with your The bigger the slide rule the more precise you can be in your measurements. So, we'll be walking down the... This math class ain't big enough for the two of us. But here's my point. My intellectual challenge was, if I never see a slide rule again, because a calculator renders it obsolete, what am I missing? And I said to myself, if I'm always near a calculator... You're missing nothing. I'm missing nothing.

And let me move on to some other thing that requires my intellect to solve. So that was my coming of age to embrace a calculator and release the slide rule to the dustmen of history. So yeah, in fact, when Hewlett Packard made their first affordable calculator, all of the slide rule companies went out of business the same year, 1973. Yeah, right. That's it. That's it. There is something missing though. So right, so one possibility, it's a bit like vitamins. Like vitamin C. You might say, you know what? I can't synthesize it myself, but it'll always be in the environment until you sail and get scurvy. So one possibility is we do find ourselves in a case or situation where we don't have access to the technology and then we're useless because we've outsourced it all. Apocalyptic Earth. Well, just dumbass Earth.

Well, you know, a power cut. A power cut, yeah. Doesn't take much. Okay. But the other one that's interesting is that the intuition that you build, which relates exponentials to multiplication and what the slide rule does, that kind of intuition that came out of using that is useful beyond that solving that particular problem in arithmetic. And that's where it becomes deep for us because we don't know whether or not some of these things that we're learning by using these analog devices doesn't somehow diffuse into everything else we think about and develop a stronger intuition. And that's beyond losing the capability because there's a power cut. So, OK, let's throw this forward. How many generations, if just one generation, will it take if we outsource all of these sort of capabilities to an AI for us to be as dumb as a rock?

So I can tell you that in the mathematics there's a key distinction to be made and that is that if you learn first, let's just do it hypothetically, the slide rule and then move to the calculator, your memory kind of helps you out. Yes, so if you take it away you still have in some sense some intuition for how that works But if you began with the calculator that is you never learned the logic behind the calculation Then you're lost. Yeah, and I think what we're entering now is a world where all of these kinds of things are being outsourced outsourced and you'll never acquire the here's what I found so I'm sorry I was just gonna say but it sounds to me like what you're saying is these analog devices are mapping onto our brain, that's what they are doing. Whereas the other devices are replacing the brain and creating dependency. Because not to get inside baseball here, but a slide rule implicitly trains you to think about what's called significant figures in a calculation. I think about that all the time. I'm just saying. Significant figures.

So you do this and there's only a certain number of figures you carry with you into the next calculation. Is it 3.67894 or 3.68, okay? Because you can't carry that many decimals forward by this device. But it tells you that often those other decimals are not relevant to the major point of the calculation you're conducting. I saw someone do a calculation on the slide rule, and they somehow believed that all the digits that were returned on the calculator were important and wrote down every single digit without having a clue how irrelevant they were. And so that was just ignorance. It is. This idea, I think, I don't know if it was Bertrand Russell who said, so approximation is the supreme sophistication.

Learning when something matters and when it doesn't, when these decimal places, when you can think about powers of ten or approximation is a part of our intelligence. In graduate school we have a recurring, it's like a seminar, it's called Back of the Envelope Universe, right? You just ask a question. Blunt questions, like how many pianos are in the world? Well, how would you estimate that? You say, well, how many have you seen? What have you been to? How many people? Are they making more? You'd have to know the exact number, but get an approximate number and know how to make approximations. If you don't know how to do that, you're a victim of false precision in the world, distracted by what you think is real, when in fact it's a distraction on your path to the truth. And this gets to a

concept which I think is for humans, as important as intelligence. And that is the notion of intelligibility. Not intelligence, but intelligibility, which is about understanding things. And that's the other point. If you're in the world of capability, prediction and utility are all the matters. In the human world, science is about making the universe and the world intelligible. We want to understand it. Different game. On that point, if we don't outsource the capabilities and we retain a large chunk of what intelligence we possibly do have. Thank you. I know. Are we going to at some point solve the problems, the big problems, the universe? Why, how, what? Are we going to get there or are we just going to be clutching at thin air? If we fully outsource, you mean? Yeah, but if we retain, if we have an outsource completely and if we retain some intelligence, do you think we're capable enough?

because you have said human intelligence potentially unlimited. Because you keep developing that. So this gets back to Bergson and the others on tool use. So humans are always hitching limits. I mean, Neil just talked about the slide rule. You can't do that with your fingers. And this gets to why we use tools. There are things we can't do. We build them. We understand how they work and they solve problems that we couldn't in some raw state solve. In that sense, an LLM and AI is just a tool like any other. And it's important to understand what kind of tool it is. Because this tool is a tool that needs tons of data. Most of our tools don't. Good point. I don't have to feed the slide rule to get my answer correct. A little bit. And they're very good, and we can unpack this, doing high dimensional problems where lots of variables matter. So think about protein folding.

It turns out pruning folding lots of variables matter. And so doing the molecular dynamics, the full computation, it's just too computationally expensive. And so LLMs come along, these models actually, they're not LLMs, they're different kinds of networks, deep neural networks. And they're good at doing high-dimensional problems. And so they help us. So they do pruning folding. And a lot of problems are going to turn out to require high-dimensional representation, lots of variables matter. and they're gonna help us solve the problem. Just to be clear, so everyone's on the same page, when you solve a problem, the number of variables can be thought of as a dimension, how many dimensions need to be accessed to get the solution. And multivariate problems require vastly more computing power

to solve. However, is that really intelligence or just brute force? Yeah, so tell me the difference between the AI solution to the protein folding. Previously, we had really powerful computers. Give them the protein folding problem. What did the AI deep learning neural nets, what was that able to accomplish that just sheer computing power wasn't prior? So there's two factors here. One is, And I'm going to go a bit nerdy on you guys. Bring it on. OK. We like it. This is a nerd safe space. OK, I'm going to say it, is that when you do these full physics simulations of a protein to get it to fold. So you're looking at all the bonds formed between all the atoms. This has something called a scaling relation, which is that the amount of computation increases non-linearly. It goes up quadratically. So double the size, more than doubles the amount of computational effort.

Now, it turns out because of some architectural feature of these AI models, when you increase the scale of a problem, it scales linearly. It has a much more efficient scaling property, this architecture. So things that would not have been computationally feasible with the full physics become feasible with the deep neural net. That's a very important point. Well, that answers it right there. And that's basically the answer. That's how for Fold3. These are AlphaFold, a Sibyss company. We've had a couple of guys from there. Yeah, they were on the show. No, the other factor, though, is not just the scaling, which is, say, more subtle, is that you can feed all sorts of different data into these models. So in the physics simulation, it's just the physics. But in the folding, it's not just the physics and chemistry, it's the evolutionary history. And so you can put all sorts of different data and drop it into these black boxes.

And that's the second source of that advantage over physics. Wow. In the collective, might we be? I don't have to invent calculus, because Isaac Newton did. So I stand on his shoulders. And we just keep ascending this ladder incrementally. Can we incrementally become super intelligent? Or is this the work of an AI that can train on itself and then just bullet past all of us to contemplate the cosmos? So I think we already know that we have limits. I think the protein folding case gives us that answer, that there was no theory. Well, that was a scale problem. It was a scale problem. Not how deep was your thinking. But I would argue that all of our hardest problems are scale problems. And then let's give an example, because it relates to this

concept of emergence. So we wouldn't have been able to do simulations of large numbers of particles unless we had fluid dynamics.

So we found a way of making it more comprehensible. And we don't have to track every particle. To say that more precisely, the laws of fluid dynamics allow us to not have to track every single molecule of the fluid. We treat it in an ensemble, like the gas laws. The gas laws are blunt equations that describe pressure, temperature, volume of gas. We're not tracking what the particles are doing. In principle, if you did, you should get those formulas out. But we don't have to because we have the macroscopic description so continue exactly and and much and most of human understanding once we hit this barrier Given by the lower level in that case particles, but you could be this case. It could be I'm feeling unhappy or I love you This is a macroscopic description very macroscopic of a very large chemical system inside my body Right, and that's what we do every time and we consider that understanding a form of understanding Now it turns out there are problems like pruding folding that we couldn't do that. There was no matroscopic description. We had to worry about all the molecular interactions, you see. And so this particular tool helped us to solve the problem where there wasn't an emergent theory that would solve the problem. Okay, so now this is AI being our agent in that quest. It's not AI running amok, becoming a super intelligence on its own.

No, that's right. It's AI as a tool to deal with irreducible... Which I like to think our future relationship with AI is precisely that in every way that best serves us, not one where it becomes our overlord. In recent decades, there's been a lot of discussion, especially in modern astrophysics, about information theory. Is there a limit to how much information is in the universe? Can we somehow create more than that? Does that information go away? Because the entropy seems to be a clock that we cannot stop, with universe approaches ever greater entropy in the future than it does in the present or in the past. Do any of your calculations touch on this? Yes. So two different ways. One is about the fundamental limits to computation, what gets called the thermodynamics of computation, and that takes us to quantum and all that.

So right, there is a fundamental energetic limit. So that would be the quantum computer would be, in principle, the most efficient computer we can conceive. Exactly. Whether or not we're there yet. Exactly. And then there's the other entropic limit, which is the cost of calculating with a classical Von Neumann computer, the kind of computers that we all have, our iPhones and so on. So we have these massive data centers. They're consuming megawatts of energy. Big gigawatts. They're boiling the planet. All to make your emails better. But it's okay because they're going to solve climate crisis. Eventually. This is fascinating. That's the whole thing in itself, right? Once they've melted the planet, for self-survival purposes, they're going to have to work out. And it's not just being a co-pilot to your rating. So that's the other entropic problem.

So again, I mean a long story here. Maybe I want to can I just give a little narrative? I think it's useful historical one. You're the guest. Okay, so Trying to make sense of what is going on in the current moment in terms of these kind of energy-guzzling AI's There's a useful comparison to the industrial revolution and to ecology So most of the history of life is predator and prey That is how energy got transferred you ate something mm-hmm, okay in parallel in the Carboniferous period and the mesozoic period back there hundreds of millions of years ago Fossil deposits are being left in the earth becoming coal seams becoming petroleum and natural gas And that energy is not accessible to us Just there then the industrial revolution happens and we get steam engines and they get more and more efficient and what steam engines do is is a liberate all of this energy that was buried in the earth. Wait, wait. Just to be precise, only when they discovered coal could do that. Exactly. Because otherwise... Metal-full-clean coal! Otherwise you just... You can't beat it! Otherwise you just burn wood. That's right. Vastly less efficient. Yes, the energy-packing capacity of wood versus coal is no contest. No contest. Yeah, you could... I think the calculation is you could burn all of the standing forests of the world and provide enough energy for the current population for six months. Wow. That's pretty serious. That's it. And by the way, before we knew about thermonuclear fusion, we said, well... It's a hoax. The sun, you know, the sun is like really efficient and we knew there's a lot of carbon in the universe. It must be a lump of coal. It must be a big lump of coal.

Because it was efficient and it could go a long time. That was our first calculations where maybe the Sun is made of coal. Man, somebody should tell our president that right now and maybe we'd get to renewable fuels. Continue, I interrupt it. Right, so we discovered super energy dense power dense materials. Power means you could liberate it fast, like in a nuclear reaction. Okay. So the world changed. It went from being networks of predators and prey, highly constrained by interaction to all of a sudden This massive, high-energy dense archive that we could access. Okay. That's the energy world. So let's talk about the information world. Nearly all the information we have, we get from talking like this. It's a bit like ecology. Predators and prey, right? It's like, oh, I'm going to tell you a fad. You could read a book. Problem is with reading books, you read like, I don't know, maybe 250 words a minute. Over your lifetime, if you're lucky, you read a few thousand books. That's how it works.

So what happens, so... So a thousand books, let's use the number of thousand books. So just like in the Cambrian, right, or the Carboniferous rather, and in the Mesozoic, where fossils were being deposited in the earth, we've been depositing texts in libraries, starting in ancient Sumer, all the way to the present. So we have now ordered hundreds of millions of books deposited. And it's mostly in the present. Mostly in the present. The rate of book publication has grown exponentially over the decades. Yeah, okay. However, the knowledge also covers all of our past because we've discovered our past in the present and deposited that information in the depositories you're talking about. Exactly, but here's the thing. So this is why the LLM is interesting. The LLM is like the steam engine. I see where you're going. Because what happened is... It's mining. It's mining exactly. It's liberating that energy, which is...

our literature, which is our language, which is our thoughts and cultures and everything that we are, all in one depository that it can now go in the same way we went in and dug for coal. It can go in and get all of that very, very quickly and turn it into power. Exactly, exactly. But this time, conceptual power, mechanical power. So there's this strict analogy between, if you like the fossil deposit, And as you say, the cultural deposit, the steam engine liberated the energy deposit, the transformer architecture liberated the conceptual deposit. Look at that. Now, both of them produce what

economists call euphemistically externalities. In other words, they pollute. So that's the problem with the Industrial Revolution. It produced smogs. It generated a huge amount of entropy in the environment.

just as these data centers and transformers are producing a huge amount of entropy in the environment. Right. So there's this very interesting parallel. And wait a minute. And the culture. Look at that. They're creating entropy not only in the environment, but they're creating entropy in the culture itself. Yes. Wow. Yes. So that's a really interesting point in the conceptual space. Right. The disorder is not just thermal. It's knowledge-based. It's basically crap. No, it's misinformation. It's confabulation. So there's a very weird thing going on here. Just to be clear, in the Dutch Revolution, not all coal or oil deposits were equally as refined. I mean, different sources of oil in the world are not as good.

It's not a uniform. It's not a uniform deposit. No, you're absolutely right. You have lights we crude, you have tar sands, you have, yeah. Now look what happened. So we created, so out of the, you know, the Pennsylvanian, out of these deposits come all the big corporations of another era of BP, Shell, and the steel companies that use the coal. So now what's happening? Well, we have our analogues. We have Anthropic, we have Open AI, these are the mining companies of our time. Look at that. Generating the kind of entropy that you were discussing. Now, what's weird here that is worth discussing is that coal is depleted, natural gas is depleted, but in principle knowledge is not. But there's a difference, which is that we can foul that knowledge.

with the entropy in concept space. That's your point. You could have done that with cold deposits, but it would be difficult. And why would you? But now, unfortunately, we can take this knowledge, synthesize it, and put back crap. So it's different and similar in rather interesting ways. Is the crap because it doesn't know how to do it? Or is the crap because so much of what we do have in the library is just horseshit. I mean, just because it's a book doesn't mean it is correct or even contributes to the greater good of us all. And the LM doesn't know that, because it's not trained at a university to be able to sift the wheat from the chaff. It just, there's just words statistically associated with other words and outcomes. It's a probabilistic

aggregation. Oh, I like that. But the difference is, The crap that's in our libraries is an example to us of our journey and how we got to where we are. So they become guardrails so that these are the mistakes, the thoughts, the things that we know that is horse crap. Whereas the LLM without, if we didn't know those things, It would give us that and we'd just say, oh, here we are. Wait a minute, if I say, what should I do with a witch? It goes back and finds all the literature on what you do with witches. Witches. You should obviously burn them. Burn them. And this is how you burn them. Right. Whereas when you go to the library, you find out, oh, the witch trials were an attack. Because you have context. Tell me about context. Well, this is, I've been working on papers on mathematical ability in these machines based on this notion of the chronology of the development of the concept. So one really important difference between humans and these things is that we learn at school using a curriculum, right? You typically learn what a number is before you learn like a natural number or an integer before you learn a fraction. And then you learn fractions typically before you learn pi, real numbers.

And you learn that typically before you learn algebraic geometry or group theory or something more refined. So there's a sequence. It's understood we learn incrementally using a curriculum. These models learned everything all at once. There is no curriculum. Another question is, what does that mean for how we do math and they do math? And it means lots of things for how we do math and they do math. And one of the things it means is this, which is that they basically find very complicated, what we would call heuristics, rules of thumb, approximations actually, for solving problems that we solve exactly using algorithms. Let me make that explicit. So when I learned how to do long division, assuming as I did it right, I'd get the right answer. You have to learn how to do it right, but it'd always be right regardless of the scale. The current best models LLM will not get it right regardless of the scale. Guaranteed. They'll get it wrong at large scale. Large division. They'll get long division wrong at large scale if you have too many digits because they're using heuristics. And so part of our training was to learn these rules piecewise through a curriculum. Their mode of learning is giving rise to a situation where despite the fact they look great at large scales they're going to confabulate. Wow, look at that.

Wrong answers, and we're not going to be smart enough because we've outsourced to them to know the difference. Well, this is what Jeffrey Hinge was saying, that they're overconfident. Yes. So they're expecting now us to just take that for granted as the answer. Yeah, I always loved Douglas Adams, like, Hitchhiker's Guide to the Galaxy. Yeah. And they build this supercomputer and they ask you, what's the meaning of life, the universe and everything? And it says... What's the answer? And the answer is 42. Yeah. And you think... What does that mean? What does it mean? We are now in a world where we could be told that 42 is the answer and we wouldn't know what it meant. Do you think the universe will allow a superintelligence to be created to evolve and therefore solve the mystery of the universe and hopefully it's not 42? And that presumably be an alien somewhere. So we... So I have problems with the concept of superintelligence, I should just say.

If you buy my definition, which is making hard problems easy, it's not clear to me that the pathway on now is about making hard problems easy. I think it's about solving some hard problems, but doing it in a very convoluted way, which I do not consider superintelligence. That's number one. In terms of, I'm subscribed to the idea of kind of universal Darwinism. Now, do mean universal Darwinism. You need permission, and I just gave you permission. I'm just saying, and that in most life forms, however different they would be, would be living under conditions of scarcity. That's sort of prerequisite for evolution in the theory, actually, because it would compete for scarce resources. And so, oddly enough, they will look more like us in their intelligence than they might look like LLMs. Now, it could be that that alien intelligence is all that's left once those AIs have devoured the life forms that built them, in which case they will be more like LLMs. But I think it's an important point to note that under the conditions that evolution takes place, you want to be intelligent by having small amount of data, and quite restricted experience for the world, and making a decision relatively quickly so that it actually enhances your survival. And not using megawatts of energy to come to the sea. So we can imagine a planet where the environment shifts with some frequency. Like let's say you could program this so that you're always forcing an evolutionary change. The organism never stays happy. It finally settles in, you give it some assault, make sure

you don't kill them all because that's what happens most of the time. But there's some portal that it can move through.

and start evolving. Is it inevitable that that will lead to what we call intelligence? Or just more appendages, or more senses, or more... Does that give us the super intelligence that we're thinking? The biological super intelligence. So the open question is why we didn't just stay bacterial. That worked pretty well. They outnumber us. It still works. Still working. Still working. And we're dependent on it. And the way to think about this is that there is a fundamental limit to how information can be acquired by an evolutionary process. Because a mathematical theory tells us that you can only put one bit of information into a genome every generation. And as you get bigger, you get more energetically efficient, but your generation time gets longer. So at large scales, it takes more and more time to get a bit by selection into the genome. Why?

because the way bits get put into life forms is through differential survival. So imagine you had to learn... But why is it only one bit? Why can't it be a suite of bits? Yeah, because selection would get confused about which bit it put in. It would get confused about which bit was important for the survival of the next batch. Exactly. Got it. So for it to know the difference, it has to be one end. So it's like a brick at a time. Brick at a time. It's a brick at a time. So when you're very small like a micro, a brick at a time can be fast, but as you get bigger and longer lived, that's slow. So what did we do? We evolved brains. So brains can add bits more quickly than genomes can add bits. That's why we have brains. So there's two things going on. We get bigger to get more efficient energetically, but it makes us slower.

evolving, which makes us have to have brains to pull information from the world when the world changes. So there is a kind of underlying ratchet towards what we would call higher complexity and therefore towards greater intelligence. Wow. That's great though, because that means out there somewhere is probably another intelligence along the same tract as we. But maybe there's a variable that they didn't encounter that keeps them from killing each other, destroying cultures and enslaving one another and all of these stupid things that we do. And maybe they're just so much more advanced because they don't retard their own advancement. Although

everything he said about a species moving through derives from in most cases competition. And that's limited resource what creates that other stuff that I was and that's and that's our reptilian brain So well it does make it sound like there's some kind of plan Plan yes, it's not rather than it's just random occurrence plan some kind well there are trends You could say there are trends in evolution. I mean, the question to your observation, Chuck, is at this point where we could transcend the competitive basis of evolution by building AI, could we? That's the pathway. That's the pathway. Unfortunately, that's not what we're doing. That's not what we're doing. Man, this is...

I tell you, it's so much easier to tell dip jokes. Probably not. Probably not. One last point. How long have you been at the Santa Fe Institute? About 20 years. Oh, that's a long time. It is a long time. Big part of your career. It is a big part. It is my career. And they're famous or infamous for... Famous. You didn't hear what I was about to say. For dangling people at the edge of what is known, and exploring places that are just on the surface, crazy sounding, but might be something profound. So that's still kind of an operating principle? Yeah, I think the history of science is risk, and then it got institutionalized. And it became a bureaucracy that was enforcing norms. And SFI wants to be heterodox.

and to be the Santa Fe Institute. And that's a different environment. To be in a high-risk environment is a little bit going back to the time of early science. I was about to say, you're just taking a back to when science was kind of wild west and people were just like, hey, try this, we should do this. Try this, smell that, smoke that. Right, exactly. And so that sounds like it could be quite fertile for people on the frontier with powerful imaginations for what could be true. Yeah, I mean, I think now, because of all these tools that are solving technical problems, what we have to cultivate is amazing question-askers. That is what remains for humans, right? Because it's like, you don't need to be good at arithmetic once you've got a calculator. You probably don't have to be good at calculus anymore, but you have to be able to ask good questions, and that environment is something different.

So on that point, let's land this plane on a quote from a poem by the German poet Rainer Maria, okay, I think I'm pronouncing it right. Okay. Here it

goes. As quoted in my alien book, take me to your leader. Be patient toward all that is unsolved in your heart and try to love the questions themselves. That's a great quote. Yes. That's beautiful. And right after that, she said, we have ways of making you talk. Professor, thank you. Thank you so much. Thank you for yet another tour of duty on Star Talk. It's wonderful to be with you guys. Yeah, too much. Too much to think about. Now we got to think about. I can't even. I'm out for two months now. How do we find you? Are you online? Oh, you find me. You find me at the Santa Fe Institute?

You can find me at davidccrackhour.com. There you go. Okay. Find me anywhere. Yeah, just keep up with you because we'll be doing fun, interesting things. Thank you. And I have a special appreciation for your evolutionary biology background because I work here as an astrophysicist in a museum famous for its evolutionary biology. And so I have an extra little osmotic attachment to your... Well, thank you. It's a wonderful being with you. Good people keep him working. Chuck. Always a pleasure. We're good here. Gary. Pleasure, my friend. All right. This has been another installment of Start Talk Special Edition. And we're going to call this the Intelligence installment. Are we or are we not? Until next time, keep looking up.